

DeBERTaIC: A Framework for Cyber Threat Analysis Integrating DeBERTa Model and Attack Intelligence Chain

Kai Cheng^{id}, Weidong Tang, Lintao Tan, Xin Li, and Jing Yang

Abstract—As the Internet of Things (IoT) technology advances, the Consumer Electronics (CE) sector is experiencing continuous growth, with more and more CE devices connecting to the network. This has led to an expansion of the attack surface and an increase in cyber threats, posing challenges to traditional network security defenses. Consequently, many organizations have started to use Cyber Threat Intelligence (CTI) to counter emerging cyber threats in a timely manner. However, current methods still face issues with balancing analysis accuracy and efficiency, as well as difficulties in visual presentation. To address these issues, this paper presents DeBERTaIC, a technical framework for analyzing and visualizing cyber attacks in CE networks. First, we develop a DeBERTa-BiGRU-CRF model for named entity recognition. DeBERTa's strong contextual understanding enables accurate identification of entity boundaries and extraction of key named entity information related to cyber attacks. Second, we design a DeBERTa - SVD - LightGBM model for attack-type recognition. Singular Value Decomposition (SVD) is used to extract input text features, and LightGBM is employed for rapid classification to ensure efficiency. Finally, by integrating information from the first two steps, we construct an attack intelligence chain based on the ATT&CK framework, graphically presenting attack sources, paths, targets, and types. This provides an intuitive analysis tool for decision makers to identify threats and formulate defense strategies. Experimental results confirm the effectiveness of our approach. The method used in this paper generally outperforms other models in terms of threat classification prediction accuracy and recall rate, with average values reaching 98.49% and 94.9%, respectively. Moreover, this method ensures efficient operation, providing a new perspective and tool for CE network security protection.

Index Terms—Cyber threat intelligence, DeBERTa model, attack intelligence chain, cyber threat analysis.

I. INTRODUCTION

DRIVEN by IoT technology, the CE sector is rapidly evolving and expanding. IoT systems significantly extend the capabilities and application range of CE devices by connecting sensors, actuators, appliances, and other devices to cloud servers [1]. This connectivity and intelligence not only

enhance data availability and automated control capabilities but also make applications such as smart homes and smart cities possible [2]. However, this also leads to an increase in cybersecurity threats as more and more devices are connected to the network. The domain of cyberspace is continuously expanding. At the same time, the scope of new cyber threats is broadening, increasing the likelihood of cyberattacks and exacerbating the severity of cyber risks.

Presently, the permutations of cyber assaults are increasingly variegated, with assailants employing an array of digital technologies and artificial intelligence to innovate and rapidly evolve their methods of attack. The pace of these updates and iterations is so swift that often security defenses are still in the process of being fortified when novel attack vectors emerge, imposing a significant strain on the safeguarding of network security. Furthermore, there may exist latent vulnerabilities within network apparatus or software systems [3], including the insidious zero-day vulnerabilities. Once these vulnerabilities are exploited, they can be catastrophic to network security, with affected parties often not realizing the vulnerability until after the attack has occurred.

In the face of these problems, traditional network security protection measures appear to be inadequate [4]. Although conventional methods such as real-time traffic monitoring, user authentication, and code vulnerability review can improve network security protection to a certain extent, they are difficult to cope with rapidly updated and iterative network threats. The rapid update of threat attack techniques makes it difficult for defense means to keep up with their iteration speed. In addition, some of the defense means rely on a priori knowledge, which means that in the face of unknown attacks, especially zero-day attacks, it often seems incompetent and difficult to effectively defend [5]. Attackers can circumvent these defenses, which are based on known information, through a variety of means, making the security protection system much less effective. This requires us to constantly revisit and adjust the existing network security protection strategies to adapt to the ever-changing network attacks.

Currently, in order to strengthen the weaknesses in traditional cybersecurity protection, many enterprises and organizations have begun to incorporate the use of Cyber Threat Intelligence (CTI) [6]. CTI is a type of data-based intelligence that collects, analyzes, and disseminates information about cybersecurity threats and vulnerabilities to help organizations understand potential cyber risks and take appropriate

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preventive or countermeasure measures. CTI is derived from a variety of sources, including public forums, social media, the dark Web, and various security incident reports. This information provides a broader view for organizations to observe and analyze cyber threats from multiple perspectives, helping them identify and understand complex cyber attack patterns and behaviors [7].

However, there are also technical problems with applying CTI. In order to analyze cyber threats, security experts rely on a large amount of threat information, which is usually presented in natural language and is often unstructured, making it challenging to extract valuable insights from it. Intelligence may also contain a large amount of text data, Web logs, system alerts, etc., which are in different formats and lack of uniform standards, posing great challenges for information extraction and analysis [8]. In addition, existing constructed threat recognition models have limitations in terms of efficiency, they usually use deep learning techniques, which are able to learn and extract features from a large amount of data, and then recognize and predict cyber threats, but in practical applications, it still faces the problems of long training time and low efficiency [9]. In addition, even if we are able to extract and identify threat content from unstructured threat intelligence, how to transform this information into intuitive and easy-to-understand visual attack logic is also a challenge. This requires not only a high level of technical processing skills, but also a high level of expertise among analysts in order to accurately interpret and present the attack logic, which is a considerable challenge for many organizations [10].

In view of these issues, we propose the approach in this paper, DeBERTaIC, a cyber threat analysis framework based on integrated DeBERTa model and attack intelligence chain. DeBERTa, which stands for decoding-enhanced BERT with disentangled attention [11], is a sophisticated pre-trained language representation model. It refines the attention mechanism of the bidirectional encoder representations from Transformers (BERT) [12], thereby enhancing model performance through the incorporation of decoupled attention and decoding-enhancement techniques. Our experimental approach involves leveraging the DeBERTa model and integrating a suite of methods, including bidirectional gated recurrent unit (BiGRU), conditional random field (CRF), and light gradient boosting machine (LightGBM), to achieve named entity recognition and classify attack behaviors within CTI. Furthermore, we construct an attack intelligence chain structure for CTI based on the MITRE ATT&CK framework, facilitating an intuitive and clear visualization of the attack logic.

To summarize, this research makes the following three main contributions:

- 1) We utilize the DeBERTa-BiGRU-CRF model to complete the identification of named entities in CTI, which solves the problem of the difficulty of unstructured natural language intelligence extraction.
- 2) We utilize the DeBERTa-SVD-LightGBM model to complete the classification and identification of attack behaviors in CTI, which improves the accuracy and efficiency of network attack behavior identification.

- 3) We take the MITRE ATT&CK framework as the basis, and construct the cyber threat information extracted through the above models into an attack intelligence chain in the form of a graph, which facilitates enterprises or organizations to analyze and communicate the threat intelligence in a visual form.

The remainder of this study is organized as follows: Section II introduces the related work. In Section III, we describe in detail the proposed methodology for integrating the DeBERTa model and the attack intelligence chain. Section IV conducts experiments and analyzes the results. Finally, the work is summarized and prospects for future research are provided.

II. RELATED WORK

In this section we provide a brief review of several relevant studies on threat analysis and intelligence modeling.

A. Threat Analysis Related Work

Cyber threat analysis is a field that focuses on identifying, evaluating, and responding to cybersecurity threats. As cyber attacks become increasingly complex and diverse, cyber threat analysis has become a key step in protecting organizations from cyber attacks. Current research trends in cyber threat analysis include building automated cyber threat analysis tools to efficiently manage extensive datasets, identify abnormal behaviors, and predict potential attacks, as well as building knowledge graphs to organize and associate threat intelligence from different sources to better understand and analyze cyber threats. Zengy et al. [13] were inspired by the method of using item side information to model high-order connectivity in recommendation systems and applied this idea to the field of cyber threat analysis. They developed an automatic cyber threat detection system called SHADEWATCHER, which uses graph neural network technology to deeply mine high-order information in audit records to improve the efficiency and accuracy of threat detection. SHADEWATCHER also has the ability to dynamically update, which helps reduce false positives, but the system also has certain complexity and needs to be improved in terms of interpretability. Li et al. [14] suggested an automated Cyber Threat Intelligence (CTI) analysis approach named K-CTIAA. This technique extracts threat activities from unstructured CTI information using pre-trained models and knowledge graphs. K-CTIAA improves the automation level and efficiency of threat analysis, but it also relies on knowledge graphs and pre-trained models. If these models or graphs are inaccurate or incomplete, it may have a negative impact on the extraction of threat behaviors and the accuracy of analysis results. Jo et al. [15] developed a new CTI system called Vulcan. Vulcan focuses on extracting descriptive or static CTI data from unstructured text and compensates for the shortcomings of IOC-centric CTI systems in providing threat technical details and tracking frequently changing infrastructure by determining the semantic relationships between these data. However, the stability of the Vulcan system may be affected by the complexity of unstructured text. Al-Taleb and Saqib [16] introduced a combined deep learning

model integrating Convolutional Neural Networks (CNN) with Quasi-Recurrent Neural Networks (QRNN). This model aims to address the challenges of existing models in real-time classification, accuracy, and false alarm rate. Through testing on the BoT-IoT and TON_IoT datasets, the model shows superiority in network threat classification performance, but it also requires a lot of computing resources and time to complete the training and deployment of the model. Irshad and Siddiqui [17] created a system for extracting features from CTL reports to identify or profile cyber threat actors (CTAs), addressing the challenge that attackers employ various obfuscation and deception methods to obscure their identities and complicate digital forensic analysis. Additionally, a technique is outlined for extracting features from unstructured CTL reports through natural language processing (NLP) technology, followed by employing machine learning algorithms to attribute Cyber Threat Actors (CTAs). Although the computational efficiency challenges posed by prolonged training durations of deep learning models for real-time CTI updates, the DeBERTaIC framework proposed in this study introduces a novel cyber threat analysis pipeline. By integrating DeBERTa pre-trained models, this architecture achieves dual objectives: enhancing the accuracy of threat intelligence named entity recognition (NER) and attack behavior classification while maintaining optimal computational efficiency. Specifically, the framework leverages DeBERTa's disentangled attention mechanism and enhanced mask decoder to enable context-aware feature extraction from unstructured CTI reports, thereby addressing the critical challenge of structured attack information processing. Empirical evaluations demonstrate that this hybrid approach outperforms conventional methods in both analytical precision and operational scalability, establishing a robust foundation for proactive cybersecurity defense systems.

B. Threat Intelligence Modeling Related Work

CTI is important information recorded by security researchers regarding past or present network threats or security events. Generally speaking, CTI reports contain many domain specific terms, including special characters that require further processing. Therefore, extracting threat intelligence entities and dependency relationships, and establishing attack intelligence models based on them, is particularly important in the field of network threat analysis. At present, there are many types of technologies for visualizing attack intelligence based on network threat knowledge both domestically and internationally: Panahnejad and Mirabi [18] introduced the APT Det KC method, which was specifically designed to detect advanced persistent threat (APT) attacks. These types of attacks are characterized by their stealthy nature and long-term infiltration into targeted systems. The APT Det KC method proposed by them aimed to address the challenges posed by APT attacks, leveraging certain algorithms and techniques to identify the tell-tale signs of such sophisticated and persistent malicious activities. Their work provided new insights into the field of network security, especially in the context of detecting and mitigating the risks associated with APT attacks that often go unnoticed for extended periods,

potentially causing significant damage to an organization's critical infrastructure and sensitive data. However, APT Det KC relies on data preprocessing steps to reduce the amount of processed data, which may require a deep understanding of the data and accurate preprocessing methods. Although this method performs well on the KDD Cup 99 dataset, its generalization ability to other datasets or the real world needs further validation and improvement. Li et al. [19] proposed the DeepAG algorithm to address the challenges posed by multi-step attacks, such as Advanced Persistent Threats (APTs), to network security. This algorithm employs system logs to identify threats and forecast attack paths, and it utilizes a transformer model to detect APT attack sequences and construct attack graphs, illustrating the potential attack paths that adversaries might follow to compromise the network. However, despite achieving reduced complexity in detection, DeepAG still demands substantial computational power and specific operating environment and configurations. Moreover, this study assumes that DeepAG has not been targeted by malicious users, nor does it contemplate the use of adversarial or tampered samples to compromise its detection capabilities. Stergiopoulos et al. [20] proposed THREE TRACE, a threat method for automatically analyzing attack graphs and prioritizing the handling of large-scale complex networks. By using GraphSAGE (an inductive graph neural network) to learn the roles of each benign entity in the data source graph, anomaly based detection at the system entity level can be achieved. However, despite conducting robustness tests against specific evasion attacks, there may still be other types of adaptive attacks (such as data poisoning attacks and graph backdoor attacks) that could affect the system. Rose et al. [21] proposed the IDERES method, which combines network traffic analysis, machine learning, and attack graph theory, to detect and respond to tampering attempts and suspicious network transactions targeting IoT devices, improving the detection capability of known and unknown attacks. However, this approach lacks the aggregation analysis of multiple attack reports, resulting in poor integrity of attack attempts. In this study, DeBERTaIC establishes an attack intelligence chain derived from CTI reports, facilitating cross-domain information aggregation to address the challenge of structured processing for attack-related data and thereby enhancing the accuracy of attack identification.

III. METHOD

A. Overview of the DeBERTaIC

The DeBERTaIC framework consists of three main components, first, we design the DeBERTa-BiGRU-CRF model to accomplish the task of named entity recognition in CTI. The DeBERTa model is known for its strong contextual understanding, while BiGRU further enhances the model's ability to capture sequential data, and then through the CRF layer, the model is able to recognize text entity boundaries more accurately. Through this step, we are able to extract key entities such as attackers, attack targets, attack types, etc. from the massive threat intelligence data, and label the text content to lay a solid foundation for subsequent intelligence

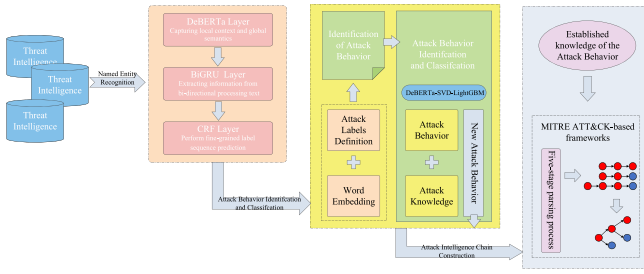


Fig. 1. Overview of the DeBERTaIC.

analysis. After that, the text content labeled in the previous section is subjected to feature extraction, and we then use the DeBERTa-SVD-LightGBM model to identify the attack types in the threat intelligence. Here, SVD is used to extract features, while LightGBM, as an efficient gradient boosting framework, is able to categorize attack types quickly and accurately. By further refining and classifying the threat intelligence information, we are able to transform the intelligence data into richer and more organized information, which provides the necessary structured data support for constructing the attack intelligence graph. Finally, we synthesize the network threat information extracted from the above model as the basis for constructing the attack intelligence chain, which uses a graphical method to display key information such as attack sources, attack paths, attack targets, and attack types. Through the connection of nodes and edges, the relationship between attackers and targets, as well as the connection between different attack types can be visualized. This visualization not only enhances the understanding of intelligence data, but also provides decision makers with an intuitive and interactive analysis tool to help them quickly identify critical threats and develop appropriate defense strategies. The specific structure of DeBERTaIC is shown in Figure 1.

B. DeBERTa-BiGRU-CRF for Named Entity Recognition

In the first part of the DeBERTaIC framework, this study designed the DeBERTa-BiGRU-CRF composite model for named entity analysis. First, DeBERTa achieves detailed processing of input data through deep coding, including word embedding and positional encoding, thereby providing an accurate representation of the absolute position information of each word in the sentence. In the encoding stage, DeBERTa uses its decomposed self-attention mechanism to capture local context and global semantics, significantly improving the coherence of word meanings. Kalyan et al. [22] also proved the superiority of DeBERTa in this task. Between the DeBERTa and CRF layers, we introduce BiGRU (Bidirectional Gated Recurrent Unit) to further enhance the performance of the model. BiGRU is able to extract information from bidirectional (forward and backward) processing of text sequences, which enables the model to capture more details of the context and long-distance dependencies. The bidirectional characteristics of BiGRU allow the model to simultaneously consider the context of the current word, thereby improving the understanding of word meaning and the ability to identify entity boundaries [23]. Next, the CRF layer utilizes the rich features

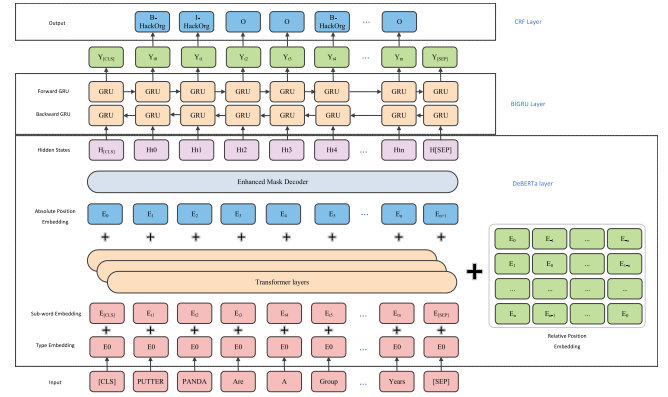


Fig. 2. The overall structure of the DeBERTa-BiGRU-CRF model.

extracted by DeBERTa and BiGRU for fine label sequence prediction. The CRF layer considers the transition probability between tags and enhances the continuity and accuracy of entity labeling [24]. This comprehensive model framework demonstrates the advantages of DeBERTa in natural language processing and significantly improves the extraction of named entities from complex text. The complete structure of this approach is illustrated in Figure 2.

Step 1: Use DeBERTa to extract feature representations from text. First, pass through the Tokenizer segmentation layer and use Byte-Level Byte Pair Encoding (BBPE) to segment the text into meaningful sub-word units. Then enter the Type Embedding layer, which can distinguish different tags and help the model get the context and relationship between sentences. Subsequently, the text enters the Sub-word Embedding layer, which converts the words into 768-dimensional vector representations to capture the semantic information and contextual relationships of the words. Finally, use the core mechanisms of DeBERTa, including the attention decoupling mechanism and the enhanced mask decoder, to further process and optimize these feature representations.

Attention decoupling mechanism: In DeBERTa's attention decoupling mechanism, each word is depicted by two distinct vectors, representing its content and its position separately. This design allows the model to consider the semantic meaning of the word and its relative position in the sentence when calculating the attention weight of the word pair. Specifically, the attention weight of a word pair can be calculated by the decoupling matrix of content and position, as shown in (1):

$$A_{i,j} = \{X_i, Y_{ij}\} \times \{X_i, Y_{ji}\}^T = X_i X_j^T + X_i Y_{ij}^T + Y_{ji} X_j^T + Y_{ji} Y_{ji}^T \quad (1)$$

where X is the content vector and Y is the relative position vector of i relative to j . Formula (1) shows that the attention weight is ultimately composed of the relationship mapping of four parts: content to content, content to position, position to content, and position to position.

Enhanced Masked Decoder: Both relative and absolute positions are crucial in the training and prediction of the model. The attention decoupling mechanism takes into account the relative position and contextual meaning, but in order to provide more complete contextual information, the model needs

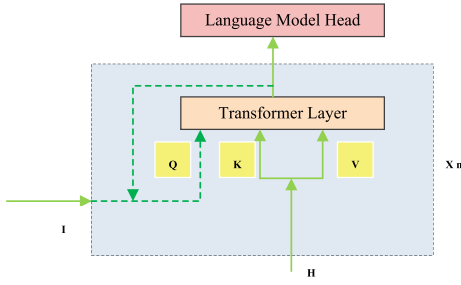


Fig. 3. The structure of Enhanced mask decoder.

to introduce absolute position information. This is particularly important in the Masked Language Modeling (MLM) task to help the model understand spatial modality. The inconsistency of the original BERT in the pre-training and fine-tuning stages may affect the performance of downstream tasks. To address these problems, DeBERTa introduces the Enhanced Masked Decoder (EMD), whose structural optimization is shown in Figure 3.

In each EMD layer, the input I represents the absolute position information, which is passed to the Q , K , and V weight matrices in the attention mechanism together with the output H from the previous Transformer Layer as an additional supplement. This design allows DeBERTa to capture relative position and word content information in all Transformer layers, while using the absolute position information only as a supplement without interfering with the model's learning process. Through such an optimized structure, DeBERTa can more effectively utilize the position information of short texts, thereby enhancing the model's performance during training and representation learning.

Step 2: Capture long-term dependencies in text sequences and extract sequence information. GRU is an improved Recurrent Neural Network (RNN) that solves the problems of gradient vanishing and insufficient long-term dependency learning ability that may occur in traditional RNN when processing long sequences [25]. In comparison with the Long Short-Term Memory network (LSTM) model, GRU simplifies the model structure and merges the input gate, output gate, and forget gate into an update gate and a reset gate, which makes the model training time shorter, with fewer parameters and a simpler structure, thus saving training costs.

When data enters the GRU, it is the update gate that determines which data can flow to the next moment:

$$\zeta_t = \sigma(W_z \cdot [h_{t-1}, x_t]) \quad (2)$$

where, σ represents the sigmoid function, the symbol $\cdot$ represents the arithmetic relation of dot product, W_z is the weight matrix, h_{t-1} is the hidden state of the previous moment, and x_t is the input information of the current moment;

The reset gate determines which past data needs to be lost:

$$r_t = \sigma(w_r \cdot [h_{t-1}, x_t]) \quad (3)$$

$$h_t = \tanh(w \cdot [r_t^* h_{t-1}, x_t]) \quad (4)$$

$$h_t = (1 - z_t) * h_{t-1} + z_t^* h_t \quad (5)$$

Among them, σ represents the sigmoid function, and the symbol $\cdot$ represents the dot product operation relationship. h_t , z_t , r_t determine the hidden state's output.

To more effectively obtain context information, we use a bidirectional GRU model, which includes a backward GRU and a forward GRU. It can obtain forward and backward data and splice the outputs of the two directions at this moment. Figure 2 BiGRU Layer shows the BiGRU network structure. The formula is as follows:

$$\vec{h}_t = GRU(x_t, \vec{h}_{t-1}) \quad (6)$$

$$\overleftarrow{h}_t = GRU(x_t, \overleftarrow{h}_{t-1}) \quad (7)$$

$$h_t = v_t \vec{h}_t + w_t \overleftarrow{h}_t + b_t \quad (8)$$

Among them, v_t and w_t are weights, and b_t is the bias.

Step 3: Consider the dependency between labels and predict the best label sequence through global optimization. Chen et al. [26] proposed combining BERT and CRF for named entity recognition. So here we use CRF as the last layer of this model. It's an undirected graph model that integrates features from both the maximum entropy model and the hidden Markov model. In the prediction task, it can account for the impact of the current prediction result on the adjacent context information or state.

Named entity recognition is usually regarded as a sequence labeling process. In the CRF layer of the model, the BIO labeling method (Begin, Inside, Outside) is used to label the start, inside, and non-entity parts of the entity. The BIO labeling method is a standard labeling format used to label the start, inside, and non-entity parts of the entity in the text. As a typical discriminative probabilistic undirected graph model, CRF needs to follow specific rules when performing sequence labeling. For example, the O label cannot be followed by the I label, but can only be followed by the B label or the O label; another example is that the I-HackOrg label cannot be directly followed by the B-OffAct label. CRF can effectively model the dependency between label sequences, thereby obtaining the optimal global solution.

Given the observation sequence $C = (c_1, c_2, \dots, c_n)$, the state sequence $Y = (y_1, y_2, \dots, y_n)$, $Z(c)$ is a normalization term, f_k is the feature function, W_k is the weight of the feature function, and $p(y|c)$ the conditional probability is:

$$p(y|c) = \frac{1}{z(c)} \exp \sum_{k=1}^K w_k f_k(y, c) \quad (9)$$

$$z(c) = \sum_y \exp \sum_{k=1}^K w_k f_k(y, c) \quad (10)$$

Throughout the training phase, CRF can calculate the probability of X under this condition of y by maximizing the log-likelihood function using the following formula:

$$P(y|X) = \frac{\exp(\text{score}(X, y))}{\sum_{\tilde{y} \in y_x} \exp(X, \tilde{y})} \quad (11)$$

$$L = \log(P(y | X)) \quad (12)$$

Among them, y is the actual label, y_x is X all possible label sequences, and L represents the loss function.

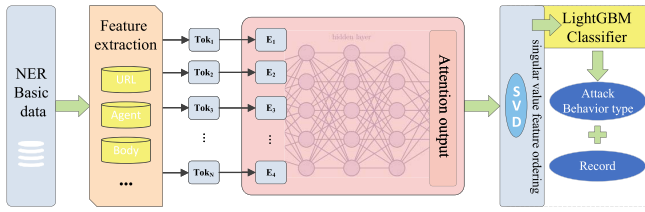


Fig. 4. The overall structure of the DeBERTa-SVD-LightGBM model.

In CRF, the Viterbi algorithm is employed to determine the globally optimal solution. y^* is the sequence with the highest score function in the sequence:

$$y^* = \arg \max_{\tilde{y} \in y_x} \text{score}(X, \tilde{y}) \quad (13)$$

C. DeBERTa-SVD-LightGBM for Attack Behavior Classification and Identification

As the second part of the DeBERTaIC framework, we designed the DeBERTa-SVD-LightGBM model for attack behavior classification and identification. In the field of threat intelligence, accurate classification of attacks is crucial, and based on the base data extracted by named entity recognition, it can quickly identify and respond to potential cyber threats, further refine and classify intelligence, and provide structured information for building attack intelligence chains. In this step, DeBERTa will be used to extract deep semantic representations of the input text. These representations capture the complex relationships and nuances in the text, provide rich features for subsequent classification tasks, and improve the model's ability to analyze and process textual data by augmenting the decoder's attention mechanism. SVD is a technique used for dimensionality reduction and feature selection, which reveals the intrinsic structure of the data [27]. With SVD, the depth representation generated by DeBERTa can be downsampled while maintaining the stream shape structure of the data. The stream shape constraint ensures that the geometric structure of the data are preserved during the feature selection process, thus helping to extract features that are more meaningful for the classification task. The LightGBM framework effectively handles classification challenges through gradient boosting techniques and tree learning algorithms. After feature selection, the filtered features are trained using LightGBM to construct a classification model. The high efficiency of LightGBM allows it to process large amounts of data quickly, enabling accurate classification of attacks and recording knowledge of novel attacks in the threat intelligence domain, thus improving the efficiency and effectiveness of security protection [28]. The specific process is shown in Figure 4.

DeBERTa model significantly improves the model's ability to understand the text by introducing decoding enhancement mechanism and decoupling attention mechanism in extracting the depth representation of the input text. Firstly, in the preprocessing stage, DeBERTa splits the input text, adds special tokens, and generates corresponding input IDs and attention masks. Subsequently, the text is converted into a word embedding representation and apply position encoding technology to ensure that the model can perceive the layout of

words in the text sequence. Unlike traditional BERT models, DeBERTa's self-attention mechanism is further subdivided by decoupling the attention mechanism to separate the query-key interaction and the key-value interaction, a design that grants the model greater flexibility and representational power. At the encoder layer, the input passes through multiple layers of Transformer encoders, each containing self-attention and feed-forward neural networks, where DeBERTa's self-attention mechanism is realized through the decoupled attention mechanism, enhancing the model's learning of different parts of the representation. In particular, DeBERTa adds a decoding enhancement layer on top of the encoder, an innovative design that helps the model to better capture long-distance dependencies, which is one of its main differences from other models such as BERT. Depending on the downstream task, DeBERTa's output may undergo further processing, such as a linear layer or a specific activation function, to suit the task requirements. By training based on a vast quantity of labeled data and updating the model parameters using the back-propagation algorithm, DeBERTa is able to minimize the loss function, thus showcasing superior capabilities in handling diverse natural language processing tasks, with text classification being a notable example.

The application of SVD in streaming constrained feature selection is a fine-grained process that reveals the intrinsic structure of the data by decomposing the data matrix into three core components. SVD is applied to the data matrix, which is decomposed into a left singular vector, a singular value diagonal matrix, and a right singular vector [27]. In this case, the right singular vector corresponds to the main directions of change in the data and the singular values measure the importance of these directions. By selecting the corresponding right singular vector with the largest singular value, we are able to identify the features that contribute the most to the flow structure of the data and achieve feature downscaling and selection. This step is the core strength of SVD in feature selection, which not only diminishes the data's complexity but also retains the most significant insights within the data. Moving forward, the selected attributes are utilized to formulate a fresh, compact feature matrix, which then underpins subsequent data analysis and modeling endeavors.

SVD can identify the features that contribute the most to the data manifold structure, achieving feature dimensionality reduction and selection. It not only reduces the dimensionality of the data but also retains the most valuable information in the data.

The working principle of SVD is to decompose any input matrix into the product of three matrices:

$$A = U \Sigma V^T \quad (14)$$

where, A is an $m \times n$ matrix, U is an $m \times m$ unitary matrix containing the eigenvectors of AA^T , Σ is an $m \times n$ diagonal matrix containing the singular values of matrix A , and V^T is an $n \times n$ unitary matrix containing the eigenvectors of $A^T A$.

LightGBM is a technique that utilizes decision tree algorithms to solve classification and regression tasks. It uses a leaf node based growth strategy in constructing the decision tree, which means that each tree splits conditionally and the

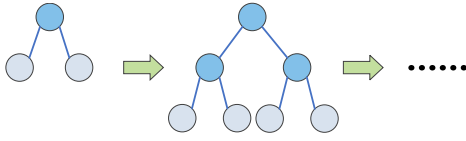


Fig. 5. Level-wise demonstration graph.

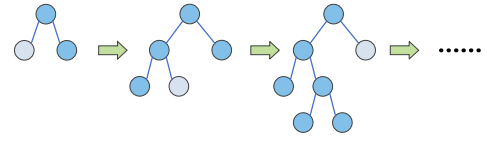


Fig. 6. Leaf-wise demonstration graph.

decision to expand new leaves is based on the level of gain. One of the notable features of LightGBM is the exclusive feature bundling, which reduces the dimensionality of the data by combining the exclusive features, thus increasing the processing speed. GBDT (Gradient Boosting Decision Tree) is an integrated model that is based on decision trees trained over multiple iterations [29]. In GBDT, decision trees are trained by iteratively fitting the negative gradient, which is synonymous with the residual error. A method of describing a GBDT model $f(x)$ as an ensemble of decision trees:

$$f(x) = \sum_{i=1}^I \gamma_i D(x; \theta_i) \quad (15)$$

where I is the number of trees in the model, γ_i is the learning rate, $D(x; \theta_i)$ is the m th decision tree, and θ_i is the tree parameters. The m th-tree is being optimized to estimate the residual error, with the objective of minimizing the loss function $\mathcal{L}$ in relation to the tree parameter θ_i .

$$\theta_i = \underset{\theta}{\operatorname{argmin}} \sum_{j=1}^N \mathcal{L}(y_j, f_{i-1}(x_j) + \gamma_i D(x_j; \theta)) \quad (16)$$

where y_j is the objective variable, $f_{i-1}(x_j)$ is the predicted value of the previous tree, and N is the number of training instances. The standard approach to optimization involves gradient descent, where the loss function's gradient is determined based on the parameters of the decision tree. However, traditional GBDT becomes highly inefficient when handling extensive volumes of data.

As shown in Figures 5, 6, the LightGBM algorithm employs a leaf node-based splitting strategy, which is more efficient compared to XGBoost's hierarchy-based splitting strategy. LightGBM's leaf-wise strategy avoids ineffective splitting of leaf nodes that do not have high gain, thus reducing unnecessary computations. In addition, LightGBM uses the Histogram algorithm to discretize continuous features to create discretized histograms. This approach allows the histogram to quickly calculate the cumulative amount of all discretized data when traversing the training data, thus accurately gauging the importance of features, the model's training efficiency and prediction accuracy are enhanced.

In attack behavior classification applications, LightGBM demonstrates higher training speed and accuracy compared to XGBoost and helps to reduce the demand for computational resources.

D. Attack Intelligence Chain for Attack Logic Representing

In the final part of DeBERTaIC, we implemented the visual representation of attack logic by constructing an attack intelligence chain. Although CTI reports provide a large

amount of information, representing the attack logic in them in a visual form is a challenging task in itself [30], and an effective approach is needed to parse and understand the complex information in these reports [31]. Therefore, in order to accurately parse the attack logic specific to a CTI report, there is a need to construct a generic framework to describe the various attack behaviours in CTIs, which will help cybersecurity experts to perform in-depth analyses. In this framework, the components of an attack behaviour are considered as nodes, while the connections and dependencies between them are represented by edges, as shown in Figure 7.

We designed a five-stage parsing process that relies on two primary data input:

Examples of behavior patterns of specific technologies obtained from the MITRE ATT&CK [32](Adversarial Tactics, Techniques, and Common Knowledge) framework describe how attackers can exploit these technologies.

MITRE ATT&CK is an open-source knowledge base developed by MITRE, an American non-profit organization, which aims to systematically classify and standardize the description of attacker's Tactics, Techniques, and Processes (TTPs) in the field of cybersecurity. By analyzing real threat intelligence and historical attack data, the framework divides the attack process into 12 tactical phases (e.g., initial access, execution, persistence, privilege elevation, etc.), and defines specific technical means under each tactic at a fine-grained level (currently containing hundreds of techniques) to form a matrix model covering different scenarios such as enterprises, mobile devices, and industrial control systems (ICS).

These two types of input data are structurally similar and are converted into two types of charts through a designed parsing process: one for a single attack technique and the other for complex attack activities involving multiple technique combinations. In short, whether the input is details of a single technology or a combination description of multiple technologies, our goal is to generate graphical representations that are easy to understand and analyze through a unified processing flow.

The construction of the attack intelligence chain employs technical templates as critical components to integrate and refine threat intelligence extracted from cyber threat intelligence (CTI) reports. These templates are systematically developed based on individual technical attack graphs derived from the MITRE ATT&CK framework, each encapsulating the behavioral patterns of a specific attack technique. Specifically, an improved alignment algorithm [33] is utilized to leverage these predefined technique templates for identifying attack activities involving multi-technique combinations within real-world CTI reports. By mapping nodes in multi-technology attack graphs to the standardized technique templates, the

where i, j are two nodes in the template graph, $i \rightarrow j$ denotes the edge consisting of these two nodes, while m, n are two nodes in the template graph, $m \rightarrow n$ denotes the edges consisting of these two nodes, and the subscript *occur* denotes the number of times the edge. At the same time, the alignment score $\Gamma_E(G_p :: G_q)$ at the edge level considers three factors: evaluate the alignment score for the nodes at both extremes of the dependency ($\Gamma(i : m)$ and $\Gamma(j : n)$), the shortest path distance between these nodes within the attack graph $C_{min}(m \rightarrow n)$, and the number of times these nodes appear in the template. If there is no direct connection between two nodes, the dependency score between them will be considered infinite. Finally, these aligned scores are normalized to fall within the range of 0 to 1. Even with the implementation of traversal algorithms, the overall computational expense of the algorithm is manageable due to the confined size of attack graphs and technical templates.

$$\Gamma(G_p :: G_q) = \frac{1}{2} \cdot (\Gamma_M(G_p :: G_q) + \Gamma_E(G_p :: G_q)) \quad (21)$$

After completing the alignment score calculation for each technical candidate, these scores are compared with the set standard threshold to screen out subgraphs that meet the technical alignment requirements. A single node in an attack graph may be matched with multiple techniques, provided that as long as the alignment scores of these subgraphs reach or exceed the threshold we set, the corresponding techniques can be identified multiple times in the graph. The results of graph alignment can integrate the general information contained in the technical template, such as alternative entities and technical means, into the corresponding links of the attack intelligence chain. This approach can embed a richer and more detailed attack intelligence chain in CTI reports, providing deeper insights and a broader perspective for analysis [34].

IV. EXPERIMENTS AND ANALYSIS

In this section, we start by introducing the experimental dataset and comparing the methods, followed by the experimental platform and module implementation details, and then analyze the DeBERTaIC framework at three levels, namely, named entity identification, attack behavior classification identification, and attack intelligence chain construction, respectively, to validate the accuracy and reliability of our approach.

A. Dataset and Comparison Methods

1) *Datasets*: To evaluate the performance of our model, we used the following models and datasets. Here is some introduction about these models and datasets:

MalwareBazaar is a comprehensive threat intelligence repository specializing in malware samples and related data. The dataset encompasses thousands of malware samples, each accompanied by detailed metadata and analysis reports. MalwareBazaar categorizes its entries into various types of information, including malware type, file hash, analysis results, and behavioral descriptions. The data include key entity

categories such as malware family, file type, infection vector, attack method, and associated indicators of compromise (IOCs).

DNRTI is a large-scale threat intelligence named entity recognition dataset covering more than 300 threat reports, totaling 182,452 words. The dataset includes 13 entity categories: security team, tool, region, time, purpose, industry, organization, method, vulnerability, hacker group, attack, sample file, and feature.

CTI-Reports is a collection of threat reports published by nlpai-lab on GitHub, containing 310,406 records. Its categories are divided into four major categories: malware, IP, URL, and hash. The malware category is further subdivided into ten subcategories, including malware.backdoor, malware.infosteal, malware.ransom, malware.unknown, malware.drop, url.normal, url.unknown, url.cncsvr, ip.unknown, and hash.

2) *Comparison Methods*: And we set up the following comparison methods to be used as evaluation experiments:

BERT: A transformer-based model that uses bidirectional attention for context understanding, crucial for tasks like NER, with pre-training and fine-tuning capabilities for NLP benchmarks.

RoBERTa [35] (Robustly Optimized BERT Pretraining Approach): An optimized version of BERT with enhanced training techniques, achieving improved performance in NLP tasks by using more data and removing the NSP objective.

GPT-2 [36] (Generative Pre-trained Transformer 2): A generative model with a unidirectional transformer architecture, adaptable for NER tasks through fine-tuning, leveraging its large-scale pre-training on diverse texts.

BERT-BiLSTM (BERT with BiLSTM-CRF): Combines BERT embeddings with BiLSTM and CRF for NER, enhancing entity identification and classification through contextual and sequence-level constraints.

DistilBERT (Distilled BERT): A distilled version of BERT that retains most of its capabilities with reduced size and increased speed, suitable for resource-limited scenarios requiring high-quality NER.

DeBERTa: Advances BERT with a disentangled attention mechanism and enhanced encoding for more accurate identification and classification of attack behaviors.

LightGBM: An efficient gradient boosting framework using decision trees for classification, suitable for large datasets and complex feature interactions in attack behavior classification.

XGBoost [37] (Extreme Gradient Boosting): A robust gradient boosting method for classification tasks, effective with structured data and enhanced through hyperparameter tuning for complex attack behavior classification.

B. Implementation Details

All experiments were executed utilizing the PyTorch version 1.9.0 framework, deployed on a high-performance workstation. This workstation was outfitted with an NVIDIA RTX 3080 graphics processing unit (GPU), an Intel Core i9-11900K octa-core processor, and a substantial 64GB of random access memory (RAM).

1) *Model Training Proces*: Our model training process mainly consists of pre-training phase and training phase:

Pre-training Stage: We use large-scale, unlabeled text data to pre-train the DeBERTa model with 50 epochs, focusing on minimizing the masked language modeling loss and next sentence prediction loss.

Training Stage: We initialize the DeBERTa-BiGRU-CRF model with pre-trained DeBERTa embeddings, and train the BiGRU and CRF layers for 50 times to minimize the named entity recognition loss, ensuring that entities such as attackers and targets are accurately extracted from threat intelligence data. Dimensionality reduction is achieved by performing SVD on the embeddings, and then the DeBERTa-SVD-LightGBM model is used for attack behavior classification, with a training time of 50 epochs to minimize the classification loss. Combining the results of these stages, the MITRE ATT&CK framework is used to build an attack intelligence chain, mapping entities and behaviors to the ATT&CK matrix, and visualizing the attack paths and connections through graphical methods to intuitively analyze attack patterns.

2) *Architecture of DeBERTa-BiGRU-CRF*: Here we describe how to implement the various layers of the DeBERTa-BiGRU-CRF.

DeBERTa Layer: We use the pre-trained DeBERTa model with hidden dimensions of 768, which is standard for many DeBERTa configurations. We do not modify the internal architecture of DeBERTa but rely on its pre-trained weights.

BiGRU Layer: We set the dimensions of the BiGRU layer as 768 (input dimension) to 256 (hidden dimension) with a dropout rate of 0.2 to prevent overfitting.

CRF Layer: We configure the CRF layer with a transition matrix that has dimensions equal to the number of labels, ensuring constraints to prevent transitions into the start and out of the stop labels. The hyperparameter for the CRF layer, such as regularization strength (if applicable), is set to a value that balances the loss and model complexity, typically adjusted through validation.

Other parameter configuration: We set the learning rate to $5e-5$, the batch size to 16, and the number of training epochs to 50. The optimizer used is AdamW with weight decay to help regularize the model.

3) *Architecture of DeBERTa-SVD-LightGBM*: Then here we describe how to implement the various layers of the DeBERTa-BiGRU-CRF.

DeBERTa Layer: We use the pre-trained DeBERTa model with hidden dimensions of 768, and rely on its pre-trained weights to obtain contextual embeddings for the input text.

SVD Layer: We apply Singular Value Decomposition (SVD) to reduce the dimensionality of the DeBERTa embeddings. We set the SVD dimensions to 768 (input dimension) reduced to 128 (reduced dimension). This helps in capturing the most important features while reducing computational complexity.

LightGBM Classifier: The LightGBM model is used for the final classification task. We configure the LightGBM parameters as follows:

- 1) Number of Estimators: 1000
- 2) Learning Rate: 0.01

3) Maximum Depth: 7

4) Number of Leaves: 31

5) Boosting Type: 'gbdt'

Input Characteristic Description: The DeBERTa-BiGRU-CRF model outputs three core feature types: entity semantic features (768-dimensional word embedding vectors with 12 BIO tag classes), entity position features (768-dimensional absolute position vectors and normalized distances), and entity association features (256-dimensional BiGRU state vectors and entity co-occurrence matrices). In NER task, the above features are combined into an 1808-dimensional vector. For multi-entity scenarios, pooling is performed, followed by dimensionality reduction to 128 dimensions using SVD to remove noise. The features are then converted into a sparse matrix format compatible with LightGBM, where entity labels are one-hot encoded to enhance the expression of class features.

Regularization: We use L1 and L2 regularization with parameters set to 0.1 and 0.2 respectively to avoid overfitting.

Other parameter configuration: We set the learning rate to $5e-5$, the batch size to 16, and the number of training epochs to 50.

As for other comparison methods, we run the original model downloaded from hugging face and set the parameters as required. In order to ensure the fairness of the experimental results, we repeated each experiment 10 times and took the average to ensure the accuracy of the experiment.

C. Experiment of Cyber Threat Named Entity Recognition

To assess the accuracy of our method in threat intelligence named entity recognition, we compared it with RoBERTa, BERT and other deep learning models commonly used in natural language processing for named entity analysis. In the Python environment, we chose the PyTorch framework with outstanding flexibility and ease of use to build a complex neural network model. After the model was built.

We chose $\text{Accuracy} = \frac{TP+TN}{TP+FP+TN+FN}$ and $\text{Precision} = \frac{TP}{TP+FP}$, $\text{Recall} = \frac{TP}{TP+FN}$, $F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$ as the evaluation metric of the model, where: TP: True Positive (entity recognized by the model correctly), FP: False Positive (entity recognized by the model incorrectly), TN: True Negative (non-entity recognized by the model correctly), FN: False Negative (entity missed by the model).

After the experiment, the results are as follows: Present method generally outperformed other models in classification prediction accuracy and recall. It outperformed other models in four indicators among three different datasets, and reached 99.03% in Accuracy of CTI-Reports data set. The experimental data on MalwareBazaar, DNRTI, and CTI-Reports datasets are respectively depicted in Figures 9, 10, and 11, as presented below. Corresponding specific values are delineated in Table I.

The results indicate that the DeBERTa-BiGRU-CRF method introduced in this study achieves a reduced training and testing duration for the named entity recognition task, and also performs well in accuracy, precision, and recall. In contrast, the existence model may have an advantage in a single indicator after training on a certain dataset (for example, in

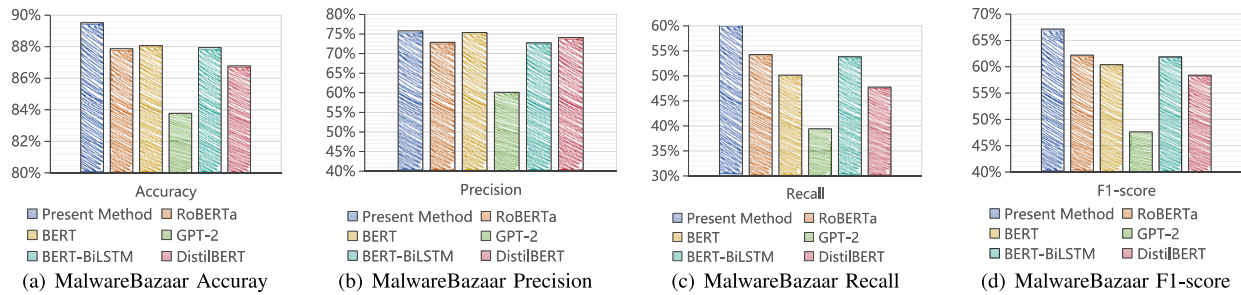


Fig. 9. Under the dataset MalwareBazaar, our model outperforms other models in all metrics in terms of accuracy, precision, recall, and F1-score (left 1, left 2, right 2, and right 1) extractions.

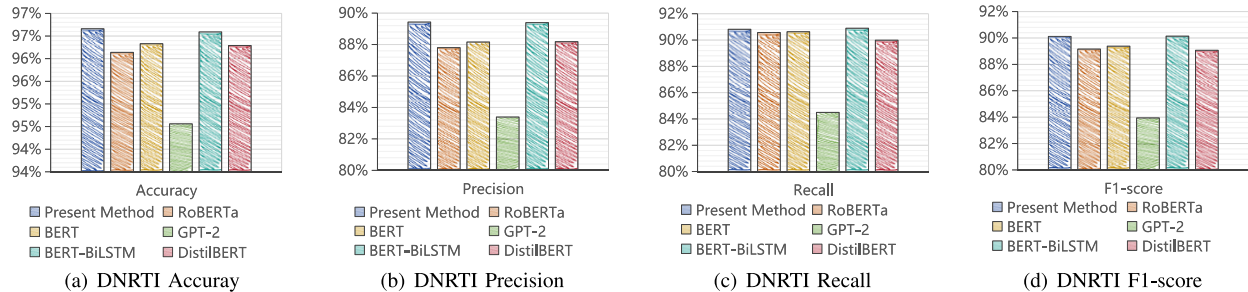


Fig. 10. Under the dataset DNRTI, our model outperforms other models in all metrics in terms of accuracy, precision, recall, and F1-score (left 1, left 2, right 2, and right 1) extractions.

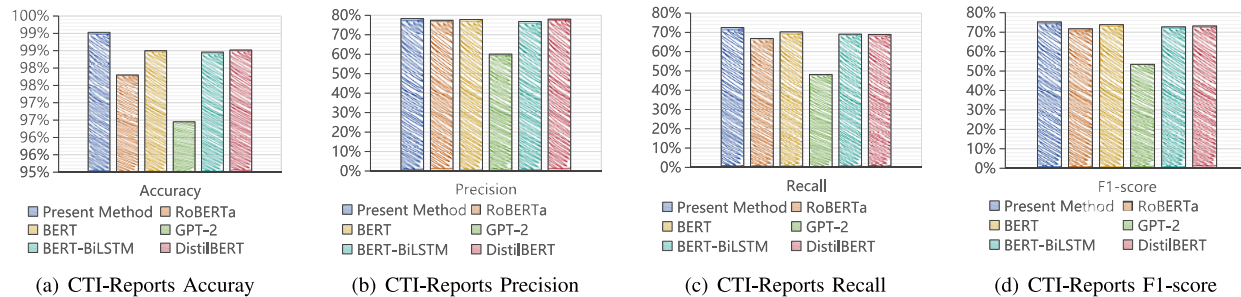


Fig. 11. Under the dataset CTI-Reports, our model outperforms other models in all metrics in terms of accuracy, precision, recall, and F1-score (left 1, left 2, right 2, and right 1) extractions.

the DRNTI dataset, BERT-BiLSTM's F1 is better than this method), but the other indicators are far behind. Overall, this study provides a stable and reliable method that has certain advantages when dealing with entity recognition tasks.

D. Experiment of Attack Behavior Classification and Identification

To assess the accuracy of our method in threat intelligence named entity recognition, we compare it with the deep learning models such as BERT, RoBERTa DeBERTa, and LightGBM, which are used today, XGBoost and other machine learning models to compare the effectiveness of attack behavior classification. We first reorganize the results of named entity recognition by integrating the extracted key information, such as user_agent, Uniform Resource Locator (URL), and message body, into feature sets for training models. These feature collections provide training data for individual models.

Afterwards, the PyTorch framework, which has outstanding flexibility and ease of use in the Python environment, is chosen as our preferred tool for building complex neural network models. In addition, two advanced machine learning libraries, LightGBM and XGBoost, were installed and integrated as the base implementations of the machine learning models for the purpose of enhancing the diversity of model comparisons. After the models were constructed, we conducted a series of experiments. All models were trained on the same training set to ensure fairness and comparability of the experiments. After the training was completed, we applied these models to the test set to predict and classify the cyber attack behaviors in it.

During the process of training models and testing, we recorded the testing and training time of each model, as shown in Table II, because the final classifier passed by this model is LightGBM, from the table, we can see that the training time of the method used in this study is much lower than that of other deep learning models, presumably because of the need to go

TABLE I
PERFORMANCE COMPARISON OF INDIVIDUAL NAMED ENTITY RECOGNITION METHODS ON VARIOUS DATASETS

Dataset	Indicator	Method					
		Present Method	RoBERTa	BERT	GPT-2	BERT-BiLSTM	DistilBERT
MalwareBazaar	Accuray	89.52%	87.87%	88.07%	83.78%	87.94%	87.72%
	Precision	75.79%	72.88%	75.89%	60.13%	72.75%	75.13%
	Recall	60.30%	54.26%	50.16%	39.42%	53.83%	47.76%
	F1-score	67.16%	62.21%	60.40%	47.62%	61.87%	58.40%
DNRTI	Accuray	96.66%	96.14%	96.33%	94.56%	96.59%	96.29%
	Precision	89.43%	87.80%	88.16%	83.39%	89.39%	88.18%
	Recall	90.82%	90.57%	90.63%	84.50%	90.90%	89.99%
	F1-score	90.12%	89.16%	89.38%	83.94%	90.14%	89.07%
CTI-Reports	Accuray	99.03%	97.80%	98.50%	96.45%	98.46%	98.52%
	Precision	78.28%	77.45%	77.85%	60.07%	76.80%	78.03%
	Recall	72.49%	66.83%	70.24%	48.17%	69.07%	68.93%
	F1-score	75.27%	71.75%	73.85%	53.47%	72.73%	73.20%

TABLE II
TRAIN&TEST TIME FOR DIFFERENT MODELS

	Present method	BERT	RoBERTa	DeBERTa	LightGBM	XGBoost
Train	28m	3h32m	4h8m	2h58m	18m	20m
Test	2m	7m	8m	5m	2m	3m

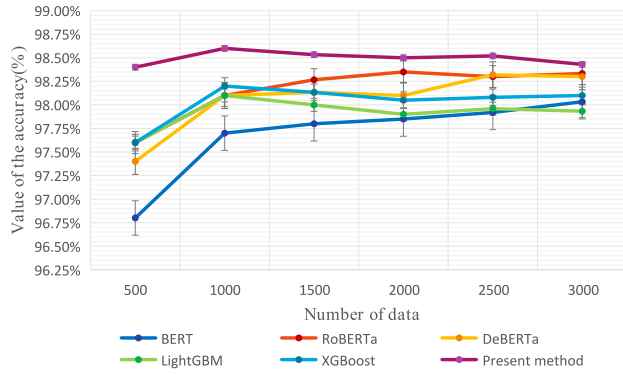


Fig. 12. Comparing the accuracy of our model with other models under data=500,1000,1500,2000,2500,3000, the accuracy of our model is higher than all other models.

TABLE III
AVERAGE OF ACCURACY AND RECALL DIFFERENT MODELS

	Present method	BERT	RoBERTa	DeBERTa	LightGBM	XGBoost
Accuracy	98.49%	97.68%	98.15%	98.05%	97.91%	98.02%
Recall	94.9%	92.32%	93.82%	93.45%	93.95%	94.03%

through the SVD layer to make feature selection to prolong the training time, but it can still illustrate the efficiency of the processing of this model.

We chose $\text{Accuracy} = \frac{TP+TN}{TP+FP+TN+FN}$ and $\text{Recall} = \frac{TP}{TP+FN}$ as the metrics for evaluating the accuracy and stability of the modeling methods. As illustrated in Figures 12 and 13, the methodology employed in this study consistently surpasses alternative models with respect to both classification prediction accuracy and recall. The method achieves average accuracies of 98.49% and recalls of 94.9%. Comparatively, the average accuracies of BERT, RoBERTa, DeBERTa, LightGBM, and XGBoost are 97.68%, 98.15%, 98.05%, 97.91%, and 98.02%, respectively. The corresponding average recalls for these models are 92.32%, 93.82%, 93.45%, 93.95%, and 94.03%, respectively. A detailed summary of these average values is provided in Table III.

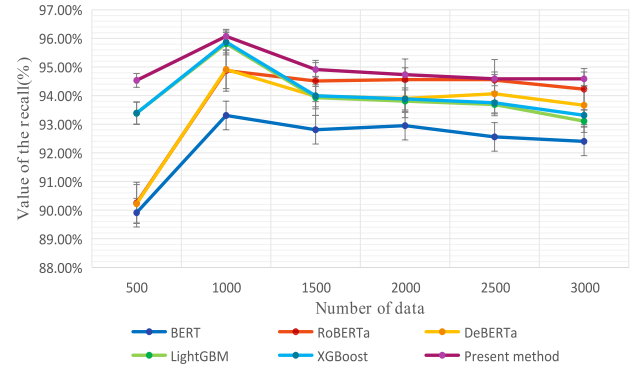


Fig. 13. Comparing the recall of our model with other models under data=500,1000,1500,2000,2500,3000, the accuracy of our model is higher than all other models.

The results show that the methods proposed in this study have shorter training test time, test time with higher accuracy and better recall. For the other methods, RoBERTa performs the best in terms of accuracy and recall, which may be due to its better generalization ability through larger dataset and optimized training process. XGBoost performs the best in terms of recall, which may be due to its ability to better tune the model to capture more positive instances when dealing with an unbalanced dataset. BERT performs slightly less than RoBERTa and XGBoost, which may be due to the fact that as an earlier model, it is not as well trained and optimized as subsequent models. DeBERTa and LightGBM perform close to each other, and they may not be as good as the Transformer-based models on some tasks, but still provide consistent and reliable results.

E. Analysis of Attack Intelligence Chain Construction

When constructing an attack intelligence chain, use a fuzzy matching algorithm based on its score to identify, analyze, and verify the techniques in the graph. Among them, the threshold for the graph alignment score has a considerable impact on the overall accuracy of the attack intelligence chain. Therefore, setting the threshold for node and graph alignment scores is a process that requires careful consideration. On the one hand, if the threshold is set too low, the system may erroneously match unrelated nodes or graphs together,

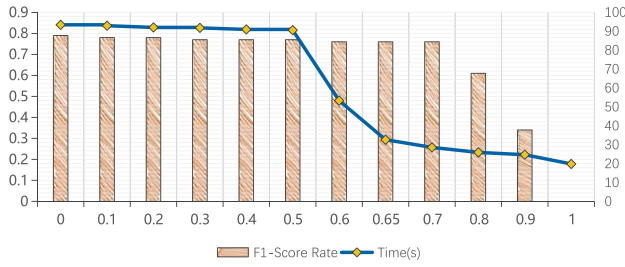


Fig. 14. F1-Score and Time performance of attack intelligence chain at different chain node thresholds (first from left, first from right) based on the chain alignment algorithm, with the chain nodes performing better in terms of overall performance at a threshold of 0.65.

which is called false positives, meaning that two entities are mistakenly considered to be matched. The system will generate a large number of candidate alignment options, which not only increases the complexity of the analysis but may also prolong the time required to complete the report. On the flip side, if the threshold is not set appropriately and is too high, some nodes or graphs that should be matched could be ignored. This situation is called false negatives, which mistakenly believe that two entities do not match. This may result in some alignment that should have been recognized being ignored, thereby reducing the overall matching accuracy. Essentially, finding the optimal threshold is crucial to maintaining the accuracy and efficiency of the attack intelligence chain. A low threshold increases the risk of incorrect matches and analysis time, while a high threshold may result in missing important matches.

When conducting experiments to evaluate the impact of different thresholds on the F1 score [38], a harmonic mean of precision and recall, providing a single metric that balances both aspects of the performance of a model. It is especially useful in cases where the cost of false positives and false negatives matters. A higher F1-score indicates better overall performance of the model in correctly classifying data. Precision measures the proportion of true positives among all predicted positives, while recall measures the proportion of true positives among all actual positives. Firstly, the data are thoroughly cleaned and standardized to ensure that the nodes and graph data involved in alignment have a unified format and meet quality standards. Next, by setting a series of different thresholds, the performance differences of the model can be observed under these thresholds. Simultaneously, cross-validation is employed to uphold the model's ability to generalize and to circumvent overfitting. This method can provide more reliable performance estimates and help us find the optimal threshold that can perform stably in different situations. Through experimental observation, it can be seen that considering the time cost and F1-score ratio, the best effect is achieved when the threshold for node alignment score is 0.65. When the time cost is significantly reduced to 32.63 seconds, the F1-score is still at 0.76, as shown in Figure 14.

When evaluating the impact of the graph alignment score threshold on model performance, the three key indicators of

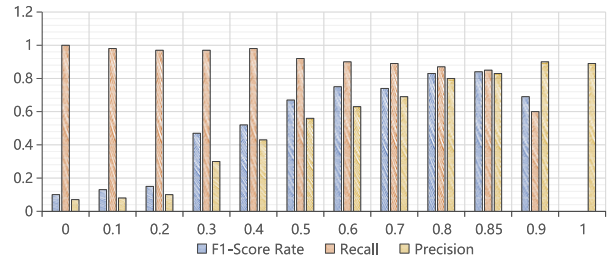


Fig. 15. Performance of the attack intelligence chain based on the chain alignment algorithm with different chain overall thresholds (from left to right, F1-Score, Recall, Precision), with a better overall chain performance at a threshold of 0.85.

accuracy, recall, and F1-score are comprehensively considered, as shown in Figure 15. This includes a comparison of model performance under different thresholds, as well as the optimal values of various indicators under each threshold. The figure shows that when the threshold for graph alignment score is set to 0.85, the model achieves a good balance in identifying correctly matched nodes and graphs, while avoiding too many incorrect and missed matches.

V. CONCLUSION AND PROSECT

In this study, we propose a cyber threat analysis framework based on integrating DeBERTa model and attack intelligence chain named DeBERTaIC, which we accurately complete the annotation and extraction of named entities in the CTI named entity recognition task by taking advantage of the DeBERTa-BiGRU-CRF model in natural language text extraction to provide basic data for the next research. In the attack behavior classification and recognition task, our DeBERTa-SVD-LightGBM model achieves 98.49% accuracy and 94.9% recall, which significantly outperforms other deep learning models, such as BERT and RoBERTa, and dramatically reduces the training time while guaranteeing high efficiency and accuracy. In the attack logic visualization task, we combine the MITRE ATT&CK framework to build an attack intelligence chain, which provides an intuitive and interactive analysis tool for cybersecurity analysis by graphically displaying attack logic.

When this scheme is applied to CE scenario, in the face of malware attacks, the DeBERTa-BiGRU-CRF model can accurately extract entities such as malware families, infection vectors, and the models of targeted devices. For example, it can identify from the reports of the MalwareBazaar dataset that the targets of the “Mirai” botnet attacks are often CE devices such as IoT cameras. Combined with the DeBERTa-SVD-LightGBM model for attack behavior classification, it can determine its behavior pattern of spreading laterally by exploiting weak passwords. The complete attack path from “scanning ports→brute force crackin→implanting malicious code→controlling devices” can be visually presented through the attack intelligence chain. In response to phishing attacks, the framework can effectively extract entities such as phishing URLs and inducement text content. It can distinguish between phishing attacks and the spread of ordinary malware

through the attack type recognition module. For example, in the analysis of phishing emails in CTI-Reports, the model accurately identified that the attack target was the smart speaker management interface of home users, the attack type was “phishing attack”, and it was mapped to the “Credential Access” tactic through the ATT&CK framework, helping security personnel quickly locate defense nodes.

In summary, the proposed methodology advances the field by delivering a novel, high-accuracy framework for cyber threat analysis, outperforming conventional approaches and offering a structured foundation for actionable intelligence. Through systematic addressing of identified limitations and strategic engagement with emerging technological frontiers, this research aims to contribute to the broader cybersecurity landscape by enabling more proactive, context-aware defense mechanisms against evolving adversarial tactics.

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